

^{35}Ca εp decay (25.7 ms) 1999Tr04,1985Ay01

Parent: ^{35}Ca : $E=0$; $J^\pi=1/2^+$; $T_{1/2}=25.7$ ms 2; $Q(\varepsilon\text{p})=1.587\times 10^4$ 11; % εp decay=95.8 3

^{35}Ca - J^π , $T_{1/2}$: From the Adopted Levels of ^{35}Ca .

^{35}Ca - $Q(\varepsilon\text{p})$: Deduced by evaluators from mass excesses of 4777 105 for ^{35}Ca measured by 2023La09 and -18378.289 80 for ^{34}Ar from 2021Wa16. $Q(\varepsilon\text{p})$ from 2021Wa16: $Q(\varepsilon\text{p})=16280$ 200 (syst).

^{35}Ca -% εp decay: 95.8 3 derived from the renormalization of % $\Sigma\text{I}(1\text{p})$ +% $\Sigma\text{I}(2\text{p})=100.6$ in 1999Tr04 to 100. The original decay branching ratios in 1999Tr04: % $\Sigma\text{I}(1\text{p})=96.4$ 18 and % $\Sigma\text{I}(2\text{p})=4.2$ 3.

1999Tr04, 1998Le45: A secondary ^{35}Ca beam at 0.3 ions/s and 98% purity was produced via the projectile fragmentation of a 95-MeV/nucleon $^{40}\text{Ca}^{20+}$ primary beam impinging on a rotating ^{58}Ni target, selected using ΔE -ToF by the GANIL LISE3 spectrometer, and implanted into a 500- μm thick silicon detector sandwiched between two 500- μm thick silicon detectors for detecting β^+ particles. 3.5×10^4 ^{35}Ca ions were stopped at a depth of 300 μm with FWHM=70 μm (setting 1) and 2.5×10^4 ^{35}Ca ions were stopped at a depth of 450 μm (setting 2). ε + β^+ -delayed protons were detected by the implantation detector. γ rays were detected by three Ge detectors and two NaI detectors. Measured E_p , I_p , E_γ , I_γ , E_{2p} , I_{2p} , βp -coin, $\text{p}\gamma$ -coin. Built the decay scheme consisting of 1p-emitting states in ^{35}K , a 2p-emitting state ($T=5/2$ IAS) in ^{35}K , 1p daughter states in ^{34}Ar , and a 2p daughter state in ^{33}Cl . Deduced decay branching ratios, B(GT) and B(F), and parent ^{35}Ca $T_{1/2}$ from implant-decay correlations.

1985Ay01: ^{35}Ca isotope discovery. ^{35}Ca was produced by bombarding a natural calcium target using a 135-MeV ^3He beam from the 88-inch Cyclotron at Lawrence Berkeley Laboratory. Recoiling products were slowed down, transported, and collected on a slowly rotating catcher wheel. The ε + β^+ -delayed protons were detected using Si detector telescopes. Measured E_p , I_p , pp -coin. Built the decay scheme consisting of a 2p-emitting state ($T=5/2$ IAS) in ^{35}K , sequential 2p intermediate states in ^{34}Ar , and 2p daughter states in ^{33}Cl . Deduced ^{35}Ca $T_{1/2}$ and ^{35}Ca mass using the known members of $A=35$, $T=5/2$ sextuplets IMME.

Theoretical studies involving ^{35}Ca decay: 2003Sm02, 1991De26, 1990Br26.

 ^{34}Ar Levels

$E(\text{level})^\dagger$	$J^\pi^\dagger$
0	0^+
2091.1 3	2^+
3287.7 5	2^+
3873 3	0^+

† From the Adopted Levels.

Delayed Protons (^{34}Ar)

Particle normalization: from the renormalization of % $\Sigma\text{I}(1\text{p})$ +% $\Sigma\text{I}(2\text{p})=100.6$ in 1999Tr04 to 100.

$E(\text{p})^\dagger$	$E(^{34}\text{Ar})$	$I(\text{p})^\dagger{}^a$	$E(^{35}\text{K})^\ddagger$	Comments
1427 5	0	48.5 13	1553	
2278 [#]	2091.1	5.4 9	4520 [@]	$E(\text{p})$: unresolved $E(\text{p})_{\text{lab}}=1909\text{--}2647$ from 1999Tr04. $I(\text{p})$: 64.0% 93 of % $I(\text{p})_{\text{tot}}=8.4$ 6 from $\text{p}\gamma$ coincidences in 1999Tr04.
2278 [#]	3287.7	1.0 4	5716 [@]	$E(\text{p})$: unresolved $E(\text{p})_{\text{lab}}=1909\text{--}2647$ from 1999Tr04. $I(\text{p})$: 12.1% 49 of % $I(\text{p})_{\text{tot}}=8.4$ 6 from $\text{p}\gamma$ coincidences in 1999Tr04.
2278 [#]	3873	2.0 7	6302 [@]	$E(\text{p})$: unresolved $E(\text{p})_{\text{lab}}=1909\text{--}2647$ from 1999Tr04. $I(\text{p})$: 23.9% 79 of % $I(\text{p})_{\text{tot}}=8.4$ 6 from $\text{p}\gamma$ coincidences in 1999Tr04.
2727 13	2091.1	6.0 5	4982	
3224 [#]	2091.1	2.2 3	5493 [@]	$E(\text{p})$: unresolved $E(\text{p})_{\text{lab}}=2947\text{--}3500$ from 1999Tr04.
3592 25	0	3.0 3	3781	
3822 36	0	3.8 3	4018	
4041 71	2091.1	2.9 3	6335	
4570 48	0	2.9 3	4788	
4754 38	0	4.2 4	4977	
5018 71	0	3.9 3	5249	

Continued on next page (footnotes at end of table)

^{35}Ca ε p decay (25.7 ms) 1999Tr04,1985Ay01 (continued)Delayed Protons (continued)

$E(p)^{\dagger}$	$E(^{34}\text{Ar})$	$I(p)^{\ddagger a}$	$E(^{35}\text{K})^{\ddagger}$	Comments
5294 48	0	0.72 18	5533	
5466 48	0	0.61 15	5710	
5616 37	0	1.43 17	5865	
5834 60	0	1.40 19	6089	
6316 [#]	0	1.09 17	6585 [@]	E(p): unresolved $E(p)_{\text{lab}}=5983\text{--}6649$ from 1999Tr04.
6783 22	2091.1	3.8 2	9168 ^{&}	$E(^{35}\text{K})$: $E(p1)_{\text{lab}}=6783\text{--}22$ corresponds to $E(^{35}\text{K})=9157\text{--}23$.
7509 [#]	0	1.1 2	7813 [@]	E(p): unresolved $E(p)_{\text{lab}}=7131\text{--}7887$ from 1999Tr04.
8802 89	0	0.41 6	9168 ^{&}	$E(^{35}\text{K})$: $E(p0)_{\text{lab}}=8802\text{--}89$ corresponds to $E(^{35}\text{K})=9144\text{--}92$.

[†] From 1999Tr04. $E(p)$ is in lab frame. $I(p)$ is obtained from the number of observed proton events and the total number of implants, with simulated proton-detection efficiencies.

[‡] Evaluators deduced $E(\text{level})(^{35}\text{K})=E(p)_{\text{lab}}\times[m(p)+m(^{34}\text{Ar})]/m(^{34}\text{Ar})+S(p)(^{35}\text{K})+E(\text{level})(^{34}\text{Ar})$, with $S(p)(^{35}\text{K})=83.6\text{--}5$ (2021Wa16), $E(\text{level})(^{34}\text{Ar})$ from the Adopted Levels of ^{34}Ar , and $E(p)_{\text{lab}}$ from 1999Tr04. For a ^{35}K proton-emitting level with multiple proton branches, evaluators take the weighted average for $E(\text{level})(^{35}\text{K})$ values deduced from each proton branch. 1999Tr04 used $S(p)(^{35}\text{K})=78\text{--}20$ from 1993Au07, which causes a small difference between the original $E(\text{level})(^{35}\text{K})$ in 1999Tr04 and the deduced $E(\text{level})(^{35}\text{K})$ here.

[#] Unresolved protons.

[@] Unresolved proton-emitting levels in ^{35}K corresponding to a group of unresolved protons populating one daughter state in ^{34}Ar .

[&] $T=5/2$ isobaric analog state in ^{35}K , which also decays by $E(2p0)_{\text{lab}}=4305\text{--}26$, proton line 8 in 1999Tr04, corresponding to $E(^{35}\text{K})=9186\text{--}27$ with $S(2p)(^{35}\text{K})=4747.5\text{--}6$ (2021Wa16) and adding a $+7\text{--}keV$ correction for the difference in the recoil effect between 1p and 2p emissions (1999Tr04). $E(^{35}\text{K})=9168\text{--}23$ is obtained from a weighted average of the three $E(^{35}\text{K})$ values of 9144 92 (p0), 9157 23 (p1), and 9186 27 (2p0). Other: $E(2p0)_{\text{lab}}=4089\text{--}30$, $E(2p0)_{\text{c.m.}}=4311\text{--}40$ (1985Ay01), corresponding to $E(^{35}\text{K})=9059\text{--}41$ with $S(2p)(^{35}\text{K})=4747.5\text{--}6$ (2021Wa16). 1985Ay01 observed $E(2p1)_{\text{lab}}=3287\text{--}30$ and proposed both 2p0 and 2p1 proceed via a sequential decay mechanism with the first proton $E(p)_{\text{lab}}=2213\text{--}keV$, corresponding to an intermediate state in ^{34}Ar at 6807 keV. However, 2p1 has been ruled out in 1999Tr04 due to the nonobservation of expected $\gamma\gamma$ coincidences. 1999Tr04 also states that the observed ratio $I(2p0)/I(1p)=0.98\text{--}9$ agrees with the calculated branching ratio $I(2p)/I(1p)=1$ for the IAS (1991De26).

^a For absolute intensity per 100 decays, multiply by 0.994.

^{35}Ca ϵp decay (25.7 ms) 1999Tr04,1985Ay01**Decay Scheme**

I(p) Intensities: I(p) per 100 parent decays

